Incidence of Type 2 diabetes in England and its association with baseline impaired fasting glucose: The Ely study 1990–2000

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Abstract

Aim To determine the incidence of Type 2 diabetes and to examine the effect of different cut-points for impaired fasting glucose (IFG) on diabetes incidence.

Methods Population-based longitudinal study (1990–2000) with clinical, anthropometric and biochemical measurements, including an oral glucose tolerance test (OGTT), in 1040 non-diabetic adults aged 40–69 years at baseline. Baseline glucose status was defined as normoglycaemia < 5.6, IFG-lower 5.6–6.0 and IFG-original 6.1–6.9 mmol/l. The all-IFG group included fasting glucose values of 5.6–6.9 mmol/l.

Results The 10-year cumulative incidence of diabetes was 7.3 per 1000 person-years. Diabetes incidence was 2.4 [95% confidence interval (CI) 1.2, 4.8], 6.2 (4.0, 9.8) and 17.5 (12.5, 24.5) per 1000 person-years in those with normoglycaemia, IFG-lower and IFG-original, respectively. Compared with normoglycaemia, the age/sex-adjusted risk [hazard ratio (HR) and 95% CI] for incident diabetes was greatest in the IFG-original category (HR 6.9; 3.1, 15.2) and increased to a lesser degree in the IFG-lower (HR 2.5; 1.1, 5.7) and all-IFG categories (HR 4.1; 1.9, 8.7). When adjusted for confounding factors, the magnitude and direction of associations persisted, with HR 1.9, 4.4 and 2.9, for the categories IFG-lower, IFG-original and all-IFG, respectively.

Conclusions Diabetes incidence is more strongly related to IFG defined as fasting glucose between 6.1 and 6.9 mmol/l than to the lower category of 5.6–6.0 mmol/l, or entire range of 5.6–6.9 mmol/l. Future studies should examine the association of IFG with cardiovascular outcomes, but for diabetes risk our study supports the use of the IFG cut-point at 6.1 mmol/l.

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Keywords diabetes, impaired fasting glucose, incidence

Abbreviations ADA, American Diabetes Association; BMI, body mass index; CHD, coronary heart disease; CVD, cardiovascular disease; FPG, fasting plasma glucose; GP, general practitioner; HR, hazard ratio; IFG, impaired fasting glucose; NFG, normal fasting glucose; OGTT, oral glucose tolerance test; pyfu, person-years of follow-up; WHO, World Health Organization

Introduction

There are few data worldwide on diabetes incidence with the diagnosis being based on longitudinal biochemical glucose measurement [1–5]. In the UK there is a particular paucity of information on diabetes incidence, with no studies based on repeat oral glucose tolerance testing (OGTT), except our

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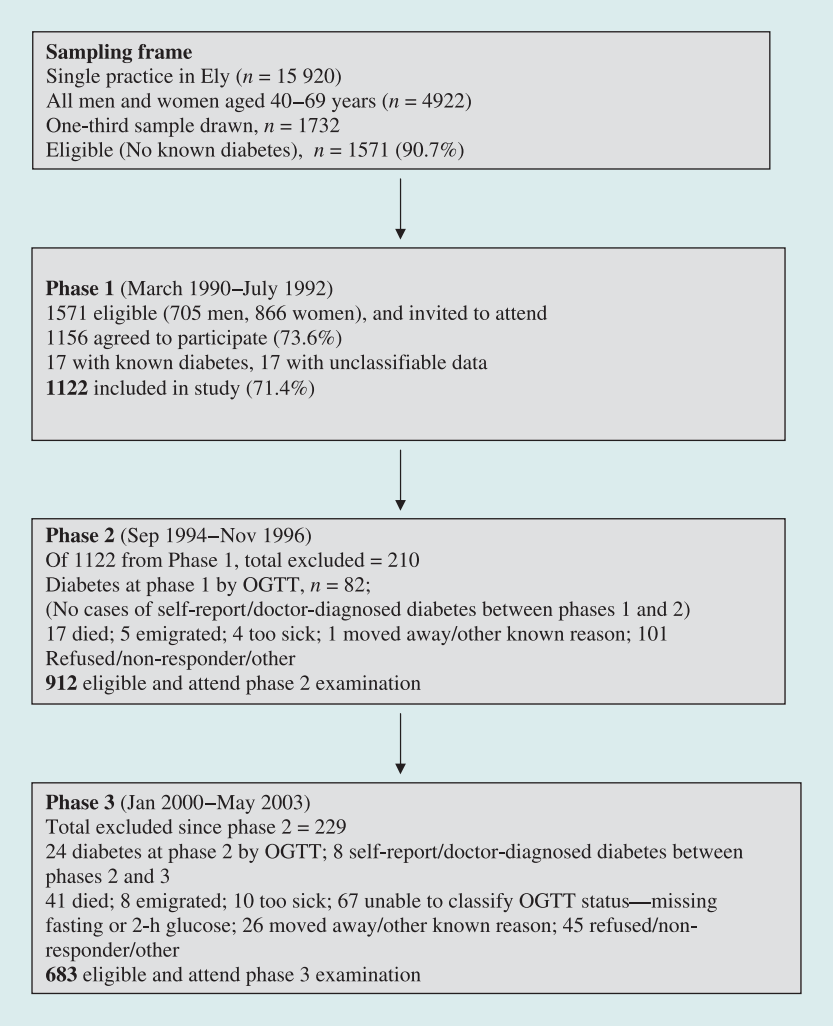

FIGURE 1 The Ely Study population at baseline and at follow-up examination.

earlier report of an incidence of 6.2 per 1000 person-years at 4.5 years of follow-up in the Ely study [6].

Recently, it has been recognized that impaired glucose regulation in the non-diabetic range is predictive of incident diabetes [1,2,5]. The category of impaired fasting glucose (IFG) was introduced in the late 1990s when the American Diabetes Association (ADA) [7] and the World Health Organization (WHO) [8] recommended that a fasting glucose of 6.1–6.9 mmol/l (110–125 mg/dl) should define a non-diabetic hyperglycaemic category. However, in 2003 the ADA recommended that the threshold for IFG be lowered to 5.6 mmol/l (100 mg/dl) [9]. This change in definition has the effect of a two- to fivefold increase in the prevalence of IFG globally [10]. However, the effect of this diagnostic lowering on new-onset diabetes has been examined in few studies [11–13] as most have tended to focus on comparisons of IFG and impaired glucose tolerance [1,2,4,5,14–17].

The aims of our study were to (i) quantify biochemically defined new-onset diabetes in the Ely prospective cohort study; (ii) examine the effect of different IFG diagnostic thresholds on the incidence of diabetes; and (iii) examine the continuous association of baseline glucose concentration with incident diabetes.

Subjects and methods

Study population

The Ely Study (Cambridgeshire, UK) was established in 1990 as a prospective study of the aetiology of Type 2 diabetes, and detailed methods were described previously [6,18]. In brief, from a sampling frame of all adults free of known diabetes registered with a single practice serving Ely, 1122 European origin adults aged 40–69 years were randomly selected (response rate 74%). The baseline examination took place between 1990 and 1992. Further follow-up occurred at a median of 4.5 years (phase 2) and 10 years (phase 3) when all individuals who were non-diabetic at baseline were invited to attend for follow-up examination (Fig. 1). At both phases 2 and 3 of the study, maximum attempts were made to trace individuals who had moved away from the Ely area, via the general practitioner (GP) and the Family Health Services Authority. Those volunteers who had moved away from Ely were invited to return for testing or, if this was not possible, the study team offered to perform testing either at the volunteer's general practice or in their own home.

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All participants gave their written informed consent and the study was approved by the Local Research Ethics Committee.

Measurements

Glucose measurement

Participants attended in the morning after an overnight fast for the measurement of fasting plasma glucose (FPG) and 2-h glucose after a 75-g glucose load for an OGTT. Blood was collected into fluoridated tubes and plasma glucose was measured in the routine laboratory using a hexokinase assay. For individuals whose tests were taken away from the study site, blood samples were spun on-site using a portable centrifuge and transferred to a −70°C freezer. For all three phases of the study the blood samples were handled in the same manner and analysed in the same laboratory.

Baseline glycaemic categories were defined as normal fasting glucose (NFG) with FPG < 5.6 mmol/l, IFG-lower as FPG 5.6–6.0 mmol/l and IFG-original as FPG 6.1–6.9 mmol/l for those with FPG < 7.0 and 2-h glucose < 11.1 mmol/l. At follow-up, incident diabetes was defined as FPG ≥ 7.0 or 2-h glucose ≥ 11.1 mmol/l or doctor diagnosis or treatment for diabetes. To enable the examination of the effect of using information from a fasting sample only, as is often the case in clinical practice, we also defined glycaemic categories using only the fasting glucose information in a second approach.

Other measurements

We performed a clinical examination including anthropometric measurements to standard protocol. The body mass index (BMI) was calculated as weight divided by the square of height (kg/m^2), and waist circumference was measured. Blood pressure was measured to standard protocol while seated. A questionnaire ascertained information on medical history, family history and physical activity from which a physical activity score was derived [19,20]. Smoking at baseline was categorized as never, ex or current smoker.

Statistical methods

We examined the distribution of baseline characteristics by glycaemic category, testing differences in proportions by χ^2 tests and in mean values across categories by analysis of variance (ANOVA) for normally distributed variables. For skewed variables (triglyceride and physical activity score) we examined the geometric mean values and 95% confidence intervals (CIs). The cumulative incidence of diabetes was calculated in rate per 1000 person-years using survival time data, with the number of participants who developed diabetes during follow-up as numerator and the total person time at risk as denominator. Cases of incident diabetes included those diagnosed on OGTT during a study visit at phase 2 or 3, and those who were clinically incident, where the date of diagnosis was taken from their medical records, and they did not receive an OGTT. If a participant declined to attend for repeat testing or was too sick to attend, the GP's case notes were reviewed to check for incidence of diagnosed diabetes. The follow-up duration up to phase 3 examination was calculated as the time between the baseline and the follow-up measurements. Data were censored for individuals who died, emigrated or were lost to follow-up for any reason. Deaths among the cohort were ascertained by information from GPs as well as by flagging the cohort with the Office for National Statistics.

We assessed the effects of baseline glucose status on incident diabetes using Cox regression with the category of normoglycaemia as the reference category. We examined age- and sex-adjusted analyses (Model 1) and further adjusted the models for potential confounders which we determined *a priori*: age, sex, BMI, waist circumference, smoking status, family history of diabetes and physical activity (Model 2). Variables age, BMI, waist circumference and physical activity were entered in models as continuous variables, while family history, sex and smoking status were categorical. We tested for the proportional hazards assumption in both observed and analytical ways and found the assumption to be valid. We tested for statistical interaction of glucose status with sex and age by using product terms and likelihood ratio tests. Maximum likelihood estimation was performed for parametric regression survival models to obtain smoothed cumulative hazard curves by glucose status.

All statistical analyses were performed using Stata/SE 8.2 (StataCorp, College Station, TX, USA). All *P*-values were based on two-sided tests, and the cut-off for statistical significance was 0.05.

Results

The study population with exclusions and losses to follow-up over time are summarized in Fig. 1.

Baseline glycaemic status

There were 1040 participants free of diabetes at baseline, with diabetes defined by OGTT criteria. Table 1 shows the age/sex distribution of baseline glycaemic status with a marked increase in the prevalence of IFG in all groups with the change in diagnostic threshold. With the original threshold of IFG of 6.1 mmol/l, 81% of the women and 67% of men were normoglycaemic, but with the lowered threshold at 5.6 mmol/l, only 47% of women and 28% of men were classified as normoglycaemic. Overall, the prevalence of IFG was 25% when based on the original definition (FPG 6.1–6.9 mmol/l), but this increased to 61% when the FPG cut-point was lowered to 5.6 mmol/l.

Across categories of normoglycaemia (FPG < 5.6 mmol/l), IFG-lower and IFG-original, there was a significant trend towards higher age, obesity (BMI and waist circumference), blood pressure and an adverse metabolic profile, as shown in Table 2.

Risk of future diabetes

Median follow-up time to final examination at phase 3 was 10.0 years (interquartile range 8.1–10.2 years). There were 61 incident cases of diabetes (8330 person-years) with an incidence

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Table 1 The distribution of fasting glycaemic status at baseline among non-diabetic men and women by age group: The Ely study 1990–2000

Fasting glycaemic status (mmol/l)	Men				Women				Total
Age (years)	40–49	50–59	60–69	All men	40–49	50–59	60–69	All women	Men + women
Number	156	155	135	446	229	201	164	594	1040
FPG < 5.6 mmol/l (%)	33.3	24.5	26.7	28.3	52.8	46.8	40.2	47.3	39.1
FPG < 6.1 mmol/l (%)	72.4	67.1	62.3	67.5	84.7	84.6	71.9	81.1	75.3
IFG-lower 5.6–6.0 mmol/l (%)	39.1	42.6	35.6	39.2	31.9	37.8	31.7	33.8	36.2
IFG-original 6.1–6.9 mmol/l (%)	27.6	32.9	37.8	32.5	15.3	15.4	28.1	18.9	24.7
All IFG 5.6–6.9 mmol/l (%)	66.7	75.5	73.4	71.7	47.2	53.2	59.8	52.7	60.9

FPG, Fasting plasma glucose; IFG, impaired fasting glucose (see Subjects and methods).

Table 2 Characteristics of non-diabetic Ely study participants by baseline fasting glucose categories

	Normoglycaemia (NFG < 5.6 mmol/l)*	IFG-lower (FPG 5.6–6.0 mmol/l)*	IFG-original (FPG 6.1–6.9 mmol/l)*	*P*-value for difference†
N (%)	407 (38.4)	376 (36.2)	257 (25.5)	
Age (years)	52.8 (7.8)	53.9 (7.5)	55.5 (7.9)	< 0.001
Men (%)	31.0	46.5	56.4	< 0.001
Current smoker (%)	19.8	21.7	17.3	—
Ex-smoker (%)	28.6	31.3	34.3	0.42
BMI (kg/m^2)	24.8 (3.6)	25.8 (4.0)	27.8 (4.6)	< 0.001
Waist circumference (cm)	79.5 (11.0)	83.6 (11.6)	90.0 (11.8)	< 0.001
Total cholesterol (mmol/l)	6.4 (1.4)	6.5 (1.3)	6.9 (1.2)	< 0.001
HDL-cholesterol (mmol/l)	1.5 (0.4)	1.5 (0.4)	1.3 (0.4)	< 0.001
Triglycerides (mmol/l)	1.1 (1.0, 1.1)	1.2 (1.1, 1.3)	1.5 (1.4, 1.6)	< 0.001
Systolic blood pressure (mmHg)	126 (16)	129 (17)	136 (16)	< 0.001
Diastolic blood pressure (mmHg)	77 (10)	79 (11)	82 (10)	< 0.001
Physical activity score (metabolic equivalent task hours per day)	11.0 (10.0, 11.7)	11.0 (10.1, 11.9)	9.6 (8.6, 10.8)	0.20

Data are arithmetic mean and SD (or geometric mean and 95% CI) or proportions.
*NFG, Normal fasting glucose; FPG, fasting plasma glucose; IFG, impaired fasting glucose.
†*P*-value for difference in continuous variables from ANOVA, or χ^2 test for difference in proportions.

rate of 7.3 (95% CI 5.7, 9.4) per 1000 person-years of follow-up (per 1000 pyfu). The cumulative incidence increased across increasing age groups with rates across ages 40–49, 50–59 and 60+ years of 5.5 (3.5, 8.8), 7.6 (5.0, 11.7) and 9.5 (6.3, 14.4) per 1000 pyfu. Incidence per 1000 pyfu was higher in men than women, respectively [10.2 (7.3, 14.1) vs. 5.2 (3.5, 7.7)]. There was no interaction between baseline age and sex, or between baseline glucose status and sex or age, and analyses were pooled for men and women, with adjustment for age and sex.

Table 3 shows that diabetes incidence rates increased across the categories of baseline glycaemia (2.4, 6.2 and 17.5 per 1000 pyfu for normoglycaemia, IFG-lower and IFG-original, respectively). Incidence in the total IFG category of 5.6–6.9 mmol/l was lower (10.6 per 1000 pyfu) than in the IFG-original group. In age- and sex-adjusted analyses, the magnitude of risk for diabetes was greater in the original IFG group than in the lower IFG group or the entire range of raised fasting glucose (5.6–6.9 mmol/l). This greater magnitude of risk persisted in the models adjusted for confounding factors. The risk association with the original IFG group was stronger than for the total IFG group which included all those with IFG-lower and IFG-original. There was no change in the magnitude or direction of the associations if we additionally adjusted for occupational social class (manual vs. non-manual occupation) and the available dietary factors (frequent vs. infrequent self-reported fresh vegetable and fruit consumption and consumption of the type of carbohydrate—rice, pasta or potatoes) in the models.

Figure 2 shows that the age- and sex-adjusted cumulative hazard for diabetes incidence over time was higher in those with original IFG compared with those in the lower IFG group and NFG. The distribution was similar for our second approach (see Subjects and methods) which used only fasting glucose information. With this approach, excluding the 2-h glucose information, the 10-year cumulative incidence would be lower at 4.8 (95% CI 3.6, 6.5) per 1000 pyfu. The corresponding incidence across categories of normoglycaemia, IFG-lower, IFG-original and total IFG (5.6–6.9 mmol/l) would be 0.6, 3.1, 14.4 and 7.6 per 1000 pyfu, respectively.

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Table 3 Ten-year cumulative incidence of diabetes and risk of diabetes by baseline fasting glucose status among individuals free of diabetes at baseline: The Ely study 1990–2000

Baseline status	Number at baseline	Incident diabetes at phase 3 (*N*)	Rate/1000 person years (95% CI)	HR Model 1* (95% CI) *P*-value	HR Model 2† (95% CI) *P*-value
Whole cohort	1040	61	7.3 (5.7, 9.4)	—	—
NFG (FPG < 5.6 mmol/l)	407	8	2.4 (1.2, 4.8)	1.0	1.0
IFG-lower (5.6–6.0 mmol/l)	376	19	6.2 (4.0, 9.8)	2.5 (1.1, 5.7) 0.032	1.9 (0.8, 4.6) 0.14
IFG-original (6.1–6.9 mmol/l)	257	34	17.5 (12.5, 24.5)	6.9 (3.1, 15.2) < 0.001	4.4 (1.9, 10.0) < 0.001
All IFG (5.6–6.9 mmol/l)	633	53	10.6 (8.1, 13.9)	4.1 (1.9, 8.7) < 0.001	2.9 (1.3, 6.3) 0.008

Diabetes is defined by oral glucose tolerance test criteria or doctor-diagnosed diabetes.
*Model 1: age and sex adjusted.
†Model 2: adjusted for baseline age, sex, body mass index, waist circumference, family history, physical activity status and smoking.
NFG, Normal fasting glucose; FPG, fasting plasma glucose; IFG, impaired fasting glucose.

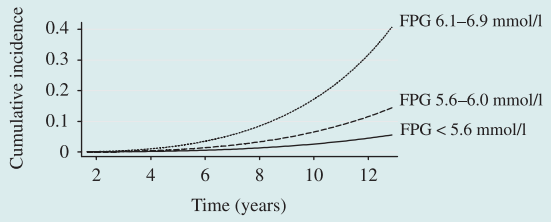

FIGURE 2 Age- and sex-adjusted cumulative hazard of incident diabetes by baseline fasting glucose status among those non-diabetic at baseline (FPG < 7.0, 2-h glucose < 11.1 mmol/l). ——, NFG; – – – –, IFG-lower; - - - -, IFG-original.

We also examined the effect of the continuous distribution of glucose in the non-diabetic range. This showed that the risk for future diabetes was markedly raised with per unit mmol/l increase in fasting glucose concentration [hazard ratio (HR) 6.4, 95% CI 3.6, 11.4, $P < 0.001$ in age/sex-adjusted analyses]. In analyses adjusted for known risk factors, the risk was elevated fourfold per mmol/l rise in glucose concentration (HR 4.2, 95% CI 2.2, 8.0, $P < 0.001$). When we examined the effect of quartiles of glucose distribution at baseline among those free of diabetes, we found that the only significant associations with new-onset diabetes were in the highest quartile of baseline glucose, at FPG ≥ 6.0 mmol/l (Table 4).

A similar proportion retained the 10-year follow-up dysglycaemic category with which they started at baseline (11.7% and 9.7% remained in the IFG-original and IFG-lower categories at follow-up, respectively). However, the category of IFG-original was more stable over time compared with that of IFG-lower, with 55.6% and 83% of individuals reverting to normoglycaemia, respectively, after 10 years of follow-up, having been dysglycaemic at baseline (Fig. 3).

Discussion

We report that the 10-year cumulative incidence of diabetes is 7.3 per 1000 pyfu in a relatively low-risk European-origin

Table 4 The association between baseline quartiles of fasting glucose in individuals without diabetes, and new-onset diabetes at 10 years of follow-up: The Ely Study 1990–2000

Quartiles of fasting glucose (range mmol/l)	Number in quartile	Incident diabetes at phase 3 (*N*)	Rate/1000 person-years (95% CI)	HR Model 1* (95% CI) *P*-value	HR Model 2† (95% CI) *P*-value
Q1 (< 5.4)	253	3	1.4 (0.5, 4.5)	1.0	1.0
Q2 (5.4–5.69)	239	9	4.6 (2.4, 8.9)	2.9 (0.8, 10.7) 0.11	2.8 (0.8, 10.6) 0.12
Q3 (5.7–5.99)	220	10	5.4 (2.9, 10.0)	3.2 (0.9, 11.7) 0.08	2.2 (0.6, 8.5) 0.27
Q4 (≥ 6.0)	322	39	16.1 (11.7, 22.0)	9.9 (3.0, 32.5) < 0.001	6.5 (1.9, 21.7) 0.003

*Model 1: age and sex adjusted.
†Model 2: adjusted for baseline age, sex, body mass index, waist circumference, family history, physical activity status and smoking.

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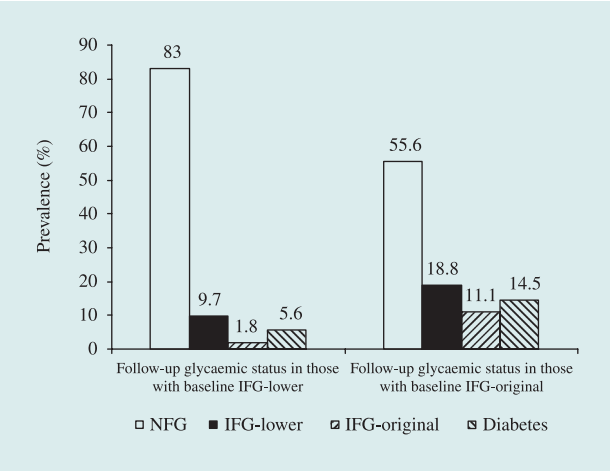

FIGURE 3 Stability of IFG categories with repeat measurement over time: The Ely study 1990–2000.

English population aged 40–69 years. This corresponds to a cumulative incidence of 5.9% and an annual incidence of 0.6%. To our knowledge, there are no other published studies of incidence in England that have quantified glucose levels biochemically with both fasting and post OGTT samples in the general population. As such, this information will help support health service planning, in particular to support delivery of the Department of Health National Service Framework for Diabetes [21,22].

Our incidence findings are of similar magnitude to other European-origin populations, such as a 10-year cumulative incidence of 8.2 per 1000 pyfu (95% CI 6.2, 10.2) among a low-risk Italian population aged 40–79 years [2]. However, our incidence rate was lower than in a Dutch population, where the 6.4-year cumulative incidence rate was 9.9% with the WHO 1999 criteria (though the rate was 6.1% with the WHO 1985 criteria) [1]. Some reasons for this difference might be that the Dutch Hoorn Study had an older population (50–75 years old) than in our study and their study population had considerably greater mean BMI [23] than ours. Among those with normoglycamia, the mean BMI in Hoorn was 26 kg/m^2 in men and 27 kg/m^2 in women, compared with the lower mean BMI of 25.4 and 24.9 kg/m^2 in Ely men and women, respectively.

Lowering the threshold of IFG to 5.6 mmol/l from 6.1 mmol/l has the effect of more than doubling its prevalence from 24.7% to 60.9% in the Ely population. However, diabetes incidence is greatest in the IFG-original group—we found incidence of 2.4, 6.2, 17.5 and 10.6 per 1000 person-years in those with normoglycaemia, IFG-lower, IFG-original and all IFG (5.6–6.9 mmol/l), respectively. This finding is similar to that recently reported in the French DESIR study, where diabetes incidence rate was one-seventh lower in the IFG-lower group compared with the IFG-original group, despite an increase in prevalence of IFG of three- to fourfold [11]. Thus, while lowering the threshold of IFG substantially increases its prevalence, the incidence of diabetes is much greater in the IFG group defined using the original criteria. Although the findings of the DESIR study were based on fasting glucose alone, our findings are based on diabetes defined by both fasting and 2-h criteria. However, in our second approach using fasting glucose information only, we also found similar associations.

We have extended the findings of previous studies by examining diabetes incidence using both fasting criteria and OGTT criteria for defining diabetes, as well as by examining incidence adjusted for known risk factors and confounders. We found that the magnitude of the association with diabetes was 2.8–4.6-fold higher for the original IFG category compared with the lower IFG category, with the scenarios of OGTT or fasting glucose criteria for diagnosis of diabetes, respectively. While most studies have focused on comparisons between IFG and IGT in relation to risk for future diabetes [1,12,15], relatively few have compared different thresholds for IFG in relation to incident diabetes. We have recently summarized the evidence for the latter associations and concluded that (i) any dichotomization of glucose concentration is arbitrary and a trade-off between sensitivity and specificity, and (ii) that there is no convincing evidence that the new lower IFG cut-point confers any benefit in identification of future diabetes that is not already identified by the original cut-point of 6.1 mmol/l [24].

We report a worse metabolic profile and greater mean levels of blood pressure among the IFG-original group compared with both the IFG-lower group and those with normoglycaemia. This highlights the importance of assessing the contribution of any baseline glycaemic status not just for future diabetes risk, but also for future cardiovascular risk. Indeed there is no support in the published literature for a greater risk for cardiovascular disease (CVD) outcomes in the IFG-lower category, as we reviewed previously [24]. Specifically, the Dutch Hoorn study reported an age- and sex-adjusted RR for CVD mortality of 1.43 (95% CI 0.79, 2.60) in the FPG group 6.1–6.9 mmol/l, but 0.63 (0.34, 1.19) in the new FPG group of 5.6–6.0 mmol/l [25]. In a study of secondary prevention in women with established coronary heart disease (CHD), Kanaya *et al.* reported that women with IFG-original (FPG 6.1–6.9 mmol/l) had increased risk of any incident CHD event (HR 1.37; 1.08, 1.74), but those with the modified 2003 ADA definition (FPG 5.6–6.9 mmol/l) were not at increased risk (HR 1.09; 0.90, 1.34), nor were women with IFG-lower [FPG 5.6–6.0 mmol/l with HR 0.90 (0.73, 1.12)] [26]. We also found that the IFG-original category was more 'stable' over time than the IFG-lower category, with 55.6% vs. 83% reverting to normoglycaemia at the 10-year follow-up measurement, respectively. These findings lend further support for retaining the original IFG cut-point rather than lowering the threshold to 5.6 mmol/l.

We found that in age- and sex-adjusted analyses the risk for diabetes increased sixfold per unit increase in baseline glucose concentration. The risk was still fourfold greater with each mmol/l increase in baseline glucose after adjustment for known risk factors and potential confounders. Thus, the risk

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for diabetes seems to increase with increasing glucose concentration in the non-diabetic range, confirming a continuous association between glucose concentration and future diabetes. When we further examined the association between quartiles of glucose distribution and diabetes, we found that there was an increasing risk across increasing quartiles, but the association was significant only in the top quartile, corresponding to a baseline glucose cut-off of ≥ 6.0 mmol/l. This finding is different from that recently reported in 13 163 younger Israeli men (aged 26–45 years), where the risk for diabetes was elevated continuously in the normal range < 5.6 mmol/l down to a level of 4.83 mol/l when compared with the lowest quintile (cut-off of 4.5 mmol/l) [27]. Potential reasons for this difference from our study could include our lower sample size and our much older population, and our inclusion of both men and women.

Some limitations of our study merit consideration. One issue is that despite our concerted effort to maximize follow-up, the participation rate at 10 years of follow-up was 72%, which, although comparable to other studies of this nature, could have potentially biased our results. When comparing baseline characteristics between those who did and did not participate in phase 3 (10 years' follow-up), participants were younger and generally healthier (lower blood pressure and fewer hypertensive, fewer obese, and lower prevalence of dysglycaemia) than non-participants. This is in line with the frequent observation of the 'healthy participant effect', which was also reported in the Dutch Hoorn study [1]. However, this would lead to an underestimation of the true cumulative incidence in the general population, and thus our results are conservative. Our population is of predominantly white-European origin, thus we were unable to describe diabetes incidence in people of other ethnic groups. In common with any study at a non-national level, the generalizability of this study is limited, so that the findings can be generalized to other similar England populations, but not necessarily to all of England, as parts of Ely are among the more affluent in England as per rankings based on the socio-economic index of multiple deprivation. However, notwithstanding the limitations, our study has notable strengths, being population based, consisting of both men and women, having fasting and postload glucose measurements to enable rigorous biochemical diagnosis of diabetes based on either fasting criteria or OGTT criteria, and a well-phenotyped sample at baseline and follow-up.

In conclusion, we report that the cumulative 10-year incidence of diabetes in a middle-aged English population is 7.3 per 1000 person-years. Diabetes incidence is greatest in those with IFG defined with the cut-off point of 6.1 mmol/l compared with the new threshold of 5.6 mmol/l or normoglycaemia at baseline. These data support the recommendation that the original cut-off of 6.1 mmol/l should be retained for defining the category of IFG. Future studies should focus efforts to quantify the risk for diabetes with baseline glucose that compare continuous distribution of glucose as well as different cut-offs for non-diabetic hyperglycaemia, and relate the risk to cardiovascular events as well as to incident diabetes.

Competing interests

None declared.

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References

1 de Vegt F, Dekker JM, Jager A, Hienkens E, Kostense PJ, Stehouwer CD *et al.* Relation of impaired fasting and postload glucose with incident type 2 diabetes in a Dutch population: The Hoorn Study. *JAMA* 2001; **285**: 2109–2113.
2 Bonora E, Kiechl S, Willeit J, Oberhollenzer F, Egger G, Meigs JB. *et al.* Population-based incidence rates and risk factors for type 2 diabetes in white individuals: the Bruneck study. *Diabetes* 2004; **53**: 1782–1789.
3 Lorenzo C, Okoloise M, Williams K, Stern MP, Haffner SM. The metabolic syndrome as predictor of type 2 diabetes: the San Antonio heart study. *Diabetes Care* 2003; **26**: 3153–3159.
4 Soderberg S, Zimmet P, Tuomilehto J, de Courten M, Dowse GK, Chitson P *et al.* High incidence of type 2 diabetes and increasing conversion rates from impaired fasting glucose and impaired glucose tolerance to diabetes in Mauritius. *J Intern Med* 2004; **256**: 37–47.
5 Wang JJ, Yuan SY, Zhu LX, Fu HJ, Li HB, Hu G *et al.* Effects of impaired fasting glucose and impaired glucose tolerance on predicting incident type 2 diabetes in a Chinese population with high postprandial glucose. *Diabetes Res Clin Pract* 2004; **66**: 183–191.
6 Wareham NJ, Byrne CD, Williams R, Day NE, Hales CN. Fasting proinsulin concentrations predict the development of type 2 diabetes. *Diabetes Care* 1999; **22**: 262–270.
7 Report of the Expert Committee on the Diagnosis and Classification of Diabetes Mellitus. *Diabetes Care* 1997; **20**: 1183–1197.
8 World Health Organization. *Definition Diagnosis and Classification of Diabetes Mellitus*. Part I. *Diagnosis and Classification of Diabetes Mellitus*. Geneva: WHO 1999.
9 Genuth S, Alberti KG, Bennett P, Buse J, Defronzo R, Kahn R *et al.* Follow-up report on the diagnosis of diabetes mellitus. *Diabetes Care* 2003; **26**: 3160–3167.
10 Borch-Johnsen K, Colagiuri S, Balkau B, Glumer C, Carstensen B, Ramachandran A *et al.* Creating a pandemic of prediabetes: the proposed new diagnostic criteria for impaired fasting glycaemia. *Diabetologia* 2004; **47**: 1396–1402.
11 Balkau B, Hillier T, Vierron E, D'hour A, Lepinay P, Royer B *et al.* Comment to: Borch-Johnsen K, Colagiuri S, Balkau B *et al.* (2004). Creating a pandemic of prediabetes: the proposed new diagnostic criteria for impaired fasting glycaemia. Diabetologia 47: 1396–1402. *Diabetologia* 2005; **48**: 801–802.
12 Qiao Q, Lindstrom J, Valle TT, Tuomilehto J. Progression to clinically diagnosed and treated diabetes from impaired glucose tolerance and impaired fasting glycaemia. *Diabet Med* 2003; **20**: 1027–1033.
13 Tai ES, Goh SY, Lee JJ, Wong MS, Heng D, Hughes K *et al.* Lowering the criterion for impaired fasting glucose: impact on disease prevalence and associated risk of diabetes and ischemic heart disease. *Diabetes Care* 2004; **27**: 1728–1734.
14 Vaccaro O, Ruffa G, Imperatore G, Iovino V, Rivellese AA, Riccardi G. Risk of diabetes in the new diagnostic category of impaired fasting glucose: a prospective analysis. *Diabetes Care* 1999; **22**: 1490–1493.

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15 Shaw JE, Zimmet PZ, de Courten M, Dowse GK, Chitson P, Gareeboo H *et al*. Impaired fasting glucose or impaired glucose tolerance. What best predicts future diabetes in Mauritius? *Diabetes Care* 1999; **22**: 399–402.
16 Meigs JB, Muller DC, Nathan DM, Blake DR, Andres R. The natural history of progression from normal glucose tolerance to type 2 diabetes in the Baltimore Longitudinal Study of Aging. *Diabetes* 2003; **52**: 1475–1484.
17 Gabir MM, Hanson RL, Dabelea D, Imperatore G, Roumain J, Bennett PH *et al*. The 1997 American Diabetes Association and 1999 World Health Organization criteria for hyperglycemia in the diagnosis and prediction of diabetes. *Diabetes Care* 2000; **23**: 1108–1112.
18 Williams DR, Wareham NJ, Brown DC, Byrne CD, Clark PM, Cox BD *et al*. Undiagnosed glucose intolerance in the community: the Isle of Ely Diabetes Project. *Diabet Med* 1995; **12**: 30–35.
19 Paffenbarger RS Jr, Blair SN, Lee IM, Hyde RT. Measurement of physical activity to assess health effects in free-living populations. *Med Sci Sports Exerc* 1993; **25**: 60–70.
20 Ainsworth BE, Haskell WL, Leon AS, Jacobs DR Jr, Montoye HJ, Sallis JF *et al*. Compendium of physical activities: classification of energy costs of human physical activities. *Med Sci Sports Exerc* 1993; **25**: 71–80.
21 Department of Health. *National Service Framework for Diabetes: Standards*. London: Department of Health 2001.
22 Department of Health. *National Service Framework: Delivery Strategy*. London: Department of Health 2003.
23 Mooy JM, Grootenhuis PA, de Vries H, Valkenburg HA, Bouter LM, Kostense PJ *et al*. Prevalence and determinants of glucose intolerance in a Dutch caucasian population. The Hoorn Study. *Diabetes Care* 1995; **18**: 1270–1273.
24 Forouhi NG, Balkau B, Borch-Johnsen K, Dekker J, Glumer C, Qiao Q *et al*. The threshold for diagnosing impaired fasting glucose: a position statement by the European Diabetes Epidemiology Group. *Diabetologia* 2006; **49**: 822–827.
25 de Vegt F, Dekker JM, Ruhe HG, Stehouwer CD, Nijpels G, Bouter LM *et al*. Hyperglycaemia is associated with all-cause and cardiovascular mortality in the Hoorn population: the Hoorn Study. *Diabetologia* 1999; **42**: 926–931.
26 Kanaya AM, Herrington D, Vittinghoff E, Lin F, Bittner V, Cauley JA *et al*. Impaired fasting glucose and cardiovascular outcomes in postmenopausal women with coronary artery disease. *Ann Intern Med* 2005; **142**: 813–820.
27 Tirosh A, Shai I, Tekes-Manova D, Israeli E, Pereg D, Shochat T *et al*. Normal fasting plasma glucose levels and type 2 diabetes in young men. *N Engl J Med* 2005; **353**: 1454–1462.

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